

Pediatric Patient Counting with Interactive Object Detection

A Mathematical Framework for Occlusion-Resistant Computer Vision

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Outline of Presentation

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The “Invisible Child” Phenomenon

- **Systemic Blindness:** Electronic health records (e.g., LHIMS) at KATH only capture the primary patient.
- **Cultural Variable:** Infants in Ghana are typically carried on their mothers' backs in crowded Outpatient Departments (OPDs).
- **The Data Deficit:** This undocumented influx skews hospital statistics, causing dangerous delays in triage.
- **The Bottleneck:** Standard Computer Vision fails due to the **Occlusion Problem**—overlapping spatial coordinates cause the algorithm to discard the infant as visual noise.

The Mathematical Evolution of Tracking

- **Era 1: Convolutional Bounding Boxes (Ouyang, 2015):** Tracked features using layered semantic networks ($F_l \in \mathbb{R}^{H \times W \times C}$). *Flaw:* Merges perfectly occluded infants into the caregiver's box.
- **Era 2: Anchor-Free Centers (FairMOT, 2021):** Shifted to tracking the absolute geometric center pixel (x_c, y_c). *Flaw:* Loses track during complete physical occlusion.
- **Era 3: Spatio-Temporal Memory (PF-Track, 2023):** Added trajectory prediction ($\hat{V}_{t+\Delta t}$) to remember hidden objects. *Flaw:* Cannot mathematically distinguish a hidden child from a backpack.
- **Era 4: Our Framework (2026):** We inject **Anthropometric Constraints** ($C \in [4.0, 6.0]$) into center-point memory, bridging kinematics and biology.

Foundation: Visual Tracking with Fully Convolutional Networks

- **Wang, Ouyang et al. (2015):** Established that convolutional layers / possess distinct properties for feature maps $F_l \in \mathbb{R}^{H \times W \times C}$.
- **Semantic vs. Discriminative:** Top layers maximize semantic category probability, while lower layers retain fine-grained spatial discrimination.
- **Our Application:** Instead of treating the CNN as a "black box" that outputs one bounding box $B \in \mathbb{R}^4$, we utilize the lower discriminative layers to isolate distinct Head (H) and Body (B) features.

Foundation: FairMOT (Zhang et al., 2021)

- **The Anchor Flaw:** Bounding box anchors overlap in crowds, causing Re-ID features to merge (Anchor Ambiguity).
- **Center-Point Heatmap:** FairMOT uses CenterNet to predict an object's center via a Gaussian heatmap:

$$Y_{x,y,c} = \exp \left(-\frac{(x - x_c)^2 + (y - y_c)^2}{2\sigma_c^2} \right)$$

- **Re-ID Extraction:** Extracts the embedding vector $E_{x,y} \in \mathbb{R}^D$ *exactly* at the peak (x_c, y_c) .

Foundation: PF-Track (Pang et al., 2023)

- **Spatio-Temporal Coherence:** Moving beyond frame-by-frame detection using "Past and Future Reasoning" on trajectory state vectors $V_t = [x_t, y_t, w_t, h_t]^T$.
- **Track Extension Mechanism:** Uses long-term trajectory predictions to estimate $\hat{V}_{t+\Delta t}$, maintaining positional states of occluded objects.
- **Our Application:** We predict Euclidean kinematics $d = \sqrt{(\hat{x}_{t+1} - x_{t+1})^2 + (\hat{y}_{t+1} - y_{t+1})^2}$ to cache geometric footprints, preventing double-counting.

Literature Pillar IV: Anthropometric Patient Identification

Foundation: Determining the Canon of Body Proportions (Domljan et al., 2024)

The Computational Translation: Standard AI relies on absolute box height (h_b), failing on short adults due to perspective distortion. Domljan et al. established that the *Head-to-Body Canon* (C) is a scale-invariant biological constant. We translate this to spatial coordinates:

$$C = \frac{\text{Body Height (BH)}}{\text{Head Length (HL)}} \approx \frac{h_b}{h_h}$$

The Biological Thresholds (95% Confidence Intervals):

- **Toddler Signature (2-4 yrs):** $C \in [5.3, 6.1]$
- **Adolescent Signature (14-15 yrs):** $C \in [7.4, 7.9]$
- **Adult Signature:** $C \geq 7.5$

The Algorithmic Override:

For any subject with pixel height $h_b < \tau$, the condition $\forall(h_b) : C > 7.0 \implies$ Adult mathematically neutralizes the "short adult" false positive, anchoring the AI in established biomechanics.

Synthesis: The "One Body, Two Identities" Logic

By merging these pillars, our proposed framework operates as follows:

- 1 **Detect:** Isolate independent Head (H) and Body (B) features (Ouyang).
- 2 **Trigger:** If one Body shape bounds two distinct Heads ($h_1, h_2 \in H$), trigger spatial evaluation.
- 3 **Evaluate Spatial Displacement:** Verify $y_{h_2} < y_{h_1}$ (Vertical Displacement).
- 4 **Verify Proportion:** Apply the Allometric Canon $C = h_b/h_h \in [4.0, 6.0]$ to confirm Pediatric status, overriding height biases.
- 5 **Track:** Assign anchor-free IDs (FairMOT) and cache trajectories (PF-Track).

Conclusion

- We are bridging the gap between hospital records and physical reality in Ghanaian healthcare.
- By combining **Anthropometric Constraints**, **Anchor-free Detection**, and **Spatio-Temporal Kinematics**, we correct standard computer vision occlusion failures.
- We transform the "Invisible Child" into actionable, privacy-compliant data, optimizing hospital triage and significantly reducing patient wait times.

Thank You.

Questions?

Selected References

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